

Dalil — Devpost Submission Draft

Oct 5, 2026 · @Kwesi Eduafo

How to use this draft

Every section below maps to a field on the Devpost "Enter a Submission" page. Paste each one in once Dalil is built and working. The deadline is Oct 30, 2026, 10:00 am PT (11:00 pm Dhaka time).

- Text in **[square brackets]** is a placeholder. Replace it with real links, numbers and results before you submit.
- "What it does" and "How we built it" describe the planned product. Before submitting, delete anything you did not actually build. Judges test the demo, and the rules require the project to work as described.
- "Challenges", "Accomplishments" and "What we learned" are prompts only. Write them from your real experience after building.

Basic fields

Field	Value
Project name	Dalil
Tagline (short)	Check a Bangladeshi land deed's ownership chain before you buy.
Track	Best Apps and Agents
Code repository	https://github.com/najmunnaharhira/nebius-global-ai-hackathonv
Demo URL	[your hosted app URL]
Demo video (YouTube, public, under 3 min)	[YouTube link]
Existed before Aug 26, 2026?	No, built from scratch during the Submission Period

Inspiration

In Bangladesh, land is most families' biggest asset, and buying it is a gamble. Fake deeds, plots sold twice, seller names that don't match the khatian (record of rights) and broken ownership chains are common. Land disputes make up a large share of the country's civil court backlog.

Most buyers can't read old deeds, which mix legal Bangla, old measurement units and handwritten entries. A lawyer's first review is slow and expensive, so many people sign without a real check. We wanted to give every buyer a fast first pass that tells them what looks wrong and exactly what to verify before they pay.

[Optional: add a personal story, such as someone you know who faced a land dispute. A real story makes the Potential Impact case much stronger.]

What it does

Dalil turns a stack of land papers into a clear ownership story and a list of red flags, in a few minutes.

1. **Upload.** The buyer uploads photos or PDFs of the deed (dalil), the khatian, mutation papers and older "via" deeds.
2. **Extract.** Dalil reads each page and pulls out the key fields: seller, buyer, dag and khatian numbers, area, mouza, deed date and registry office. The buyer can review and correct any field before analysis.
3. **Rebuild the chain.** An agent links the documents into an ownership timeline, from the original recorded owner to today's seller.
4. **Flag problems.** Dalil checks the chain and highlights issues such as:
 - The seller's name doesn't match the khatian.
 - The area being sold is larger than what the seller owned.
 - The land was sold before the seller acquired it.
 - A link in the ownership chain is missing.
 - The same dag number appears in two different sales.
5. **Tell you what to check.** Dalil produces a verification checklist for the sub-registry and land office, in Bangla and English.

Dalil is a pre-check, not legal advice. Every report says so and points the buyer to a lawyer and the land office for confirmation.

How we built it

Every model call in Dalil goes through the **Nebius Token Factory** inference API, and we route each step to the NVIDIA Nemotron model that fits it.

Step	Model / service	Why this choice
Per-page field extraction and cleanup	Nemotron Nano via Token Factory	Many small, fast calls (one per page) keep the app responsive and credits low
Chain-of-title reasoning and red-flag analysis	Nemotron 3 Ultra via Token Factory	One heavy reasoning pass over all documents together
Bilingual checklist generation	[Nemotron Super or Nano] via Token Factory	Structured output from the Ultra analysis
Large uploads in the background	[Nebius Serverless Jobs, if used]	"Your report is ready" flow for big document sets
Hosting	[Nebius Serverless Endpoints / other host]	Public demo for judges

Where Token Factory helped: [Fill in after building. For example: one OpenAI-compatible API for every model let us switch between Nano and Ultra by changing one parameter, and we compared models side by side in a day.]

Stack: [frontend, backend, database, e.g. Next.js + FastAPI].

The UI shows which model handled each step, so judges can see the routing in action.

Challenges, accomplishments and what we learned

Write these after building, from what really happened. Judges notice made-up struggles. Prompts to answer:

Challenges we ran into

- How well did the model read Bangla deeds, especially old or handwritten ones? What did you do when it failed?
- How did you match the same person across documents when names are spelled differently?
- How did you keep the red-flag checks from producing false alarms?

Accomplishments we're proud of

- [A real number, e.g. "correctly rebuilt the chain on X of Y sample deed sets"]

- [The feature that worked best in the demo]

What we learned

- [What you learned about land records in Bangladesh]
- [What you learned about splitting work between Nano and Ultra]

What's next for Dalil

- Better reading of old and handwritten deeds, tested on more real-world samples (with owners' consent).
- A mobile-first version, since most buyers will photograph papers with a phone.
- A shareable report a buyer can send to their lawyer or family.
- Partnerships with lawyers and land-office help desks, so a flagged report can turn into a booked consultation.
- Support for more document types, such as porcha, DCR and land tax receipts.

Built with

nebius-token-factory, nvidia-nemotron, nemotron-nano, nemotron-3-ultra, [nebius-serverless-jobs], [nebius-serverless-endpoints], python, [your frontend framework], [your backend framework]

Only list what you actually used.

Feedback on Nebius and NVIDIA tools

This field is judged on its own for completeness, viability and impact, so make it specific. Keep a running log while you build and fill these in from it:

- **Token Factory:** What was easy (setup, API, docs)? What slowed you down (rate limits, errors, latency, missing features)? Give one concrete improvement.
- **Nemotron models:** How did Nano and Ultra compare on speed and quality for your task? How well did they handle Bangla text? Any failure examples?
- **Serverless Jobs / Endpoints (if used):** How long did deployment take? What was confusing?
- **Builder Program and credits:** Were the credits enough? Was onboarding clear?

Example of the level of detail judges reward: "Nemotron Nano extracted dag numbers correctly on [X] of [Y] printed pages but often misread handwritten Bangla digits. A Bangla-tuned vision model in Token Factory would make this use case much stronger."

Demo video script (2:45 total)

Use only synthetic sample deeds you made yourself. Use no copyrighted music; silence or royalty-free audio is safest.

Time	On screen	Voiceover
0:00–0:20	A stack of land papers, then the Dalil home screen	"In Bangladesh, buying land means trusting papers most people can't read. Fake deeds and double sales are common. Dalil checks them for you."
0:20–0:50	Uploading 4 sample documents	"I upload the deed, the khatian and two older via deeds."
0:50–1:20	Extracted fields, with a model label showing Nemotron Nano; one field corrected by hand	"Nemotron Nano on Token Factory reads every page in seconds. I can fix anything it got wrong."
1:20–2:00	Ownership timeline building, then red flags appearing	"Nemotron 3 Ultra links the documents into one ownership chain and finds the problems: here, the seller sold more land than he owned."
2:00–2:25	Bangla/English checklist, toggle between languages	"Dalil tells the buyer exactly what to verify at the sub-registry office, in Bangla or English."
2:25–2:45	Architecture slide, then the repo link	"Nano for speed, Ultra for reasoning, all on Nebius Token Factory. Dalil: a pre-check before the biggest purchase of your life."

Testing instructions for judges

1. Open [demo URL]. No login is needed. [If a login is needed: username [x], password [y].]
2. Click **Try a sample case** to load a ready-made set of synthetic documents, or upload your own from the `samples/` folder in the repo.
3. Review the extracted fields, then click **Analyze**.
4. Look at the ownership timeline, the red flags and the checklist. Use the language toggle to switch between Bangla and English.

The demo stays free and open until the Judging Period ends on Dec 15, 2026. All sample documents are synthetic; no real person's data is used.

Pre-submit checklist

- ☐ Registered on nebiusglobalaihackathon.devpost.com ("Join Hackathon")
- ☐ Repo is public with an MIT license visible in the About section
- ☐ README has setup steps and a section on Nemotron and Token Factory use
- ☐ App makes real runtime calls to the Token Factory API
- ☐ Demo URL works in an incognito window
- ☐ Video is under 3 minutes, public on YouTube, with no copyrighted music
- ☐ Every [placeholder] in this draft is replaced or deleted
- ☐ Feedback field is filled in with specifics
- ☐ Track selected: Best Apps and Agents
- ☐ Submitted before Oct 30, 10:00 am PT (11:00 pm Dhaka); aim for Oct 29